ECE6420 - Wireless IC Design

3 units – Fall 2026

Prerequisite: ECE 4430

Grading: Letter

Time and location: T/TH 2:00-3:15PM; Van Leer C241

Instructor: Jane Gu
Instructor Office: TSRB 447;
Instructor Office hours: T 3:30-4:20PM

TA: Austin Chae, schae49@gatech.edu
TA Office hours and Locations: TBD
Mondays TBD
Wednesdays TBD
Thursday TBD

Evaluation: The grade weighing is tentatively arranged as:
homework (15%), mid-term exam (30%), final exam (40%), and design project (15%).

Homework is normally distributed on Fridays and due the following Friday. All assignments, including homework, project report, etc., should be submitted though the Canvas course website. PDF format is preferred for assignment upload. It is the student's responsibility to compare your answers with solutions to make yourself clear. You are welcome to participate in the office hours should you have any questions. **Homework is the best practice and there is NO practice exam, please pay sincere effort and work on every problem by yourself.**

The empty sections in the handouts are intended for students to fill in during class as a vital part of the learning process. My handwritten notes are not intended for sharing and will not be posted on the course website, as they are not structured for distribution.

Late policies and No exceptions: late by 1 day 30% deduction, by 2 days 60% deduction; After that, no points.

Examinations: Makeup exams will NOT be offered. Students are expected to attend both the midterm and final exams as scheduled. In the event of an unavoidable circumstance, students must notify the instructor in advance to discuss appropriate arrangements.

Grading Scheme:

Letter Grade	Range
A	100% to 80%
B	< 80% to 70%
C	< 70% to 60%
D	< 60% to 50%
F	< 50% to 0%

Re-grading: If you think you deserve more points on a project report or examination question, write a short e-mail to the TA (with copy to the instructor) indicating what should be reconsidered and including a copy of your work. The TA (and/or the instructor) will review the grading, reassign points if justified, and discuss the case with you. **The due date for regarding is one week after the solutions have been posted and graded material is returned**.

This course is highly demanding. Review the grading scheme carefully and assess whether you are prepared. If you score below 80% on the evaluation quiz, you should spend additional time strengthening the prerequisites in addition to the course's heavy workload. If you do not feel ready, consider auditing the course this year and taking it next year for a stronger foundation and a better grade.

Computer Resources:

- Students need to sign agreement before using Cadence: https://eulas.ece.gatech.edu/cadence/
- ece-labsrv.ece.gatech.edu (prior system: ece-linlabXX), please refer the link https://help.ece.gatech.edu/labs/names

Course Description

I. Basic concept of RF design for wireless communications
- A. Device nonlinearity and their effects in RF systems, gain compression, intermodulation, desensitization etc.
- B. Device noise types, models, analysis and NF
- C. System sensitivity and dynamic range

II. RF Transceiver Architecture Analysis and Design
- A. Review of heterodyne and homodyne architecture
- B. Image rejection receiver, Hartley and Weaver receiver
- C. Polyphase Filter

III. Integrated Low Noise Amplifier (LNA) Design
- A. Input matching for integrated LNA
- B. Integrated LNA topologies, common gate, common source, inductive degeneration LNAs

C. LNA design examples, transformer coupled, noise cancellation LNAs

IV. Integrated Mixer Design

A. Integrated passive and active mixer design and comparison, linearity, noise analysis. Noise folding effects
B. Mixer linearization and noise improvement techniques, source degeneration, offset transconductance

V. Voltage Controlled Oscillator (VCO)

A. Review of oscillator and VCO model
B. Phase noise generation mechanism, analysis and effects
C. Quadrature signal generation

VI. Phase Locked Loop (PLL) and Components

A. Type-I and Type-II charge pump PLL
B. Dividers including static, dynamic and programmable dividers
C. Phase Frequency Detector and Charge Pump design and nonidealities
D. Phase noise contribution and analysis from individual blocks

VII. Advanced Topics: Synthesizers or Power Amplifier

Textbooks:

"RF Microelectronics" Behzad Razavi, 2nd Edition, 2012
"The Design of CMOS Radio-Frequency Integrated Circuits" Thomas Lee, 2nd Edition,2003